21/7/26
EX.NO:1

Study of various Network used in Linux and Windows.

Aim:

Study of various Network used in Linux and Windows to of Basic Neworking Commands.

1. ping:

1. ping

```
C:\Windows\system32>ping google.in

Pinging google.in [142.250.207.68] with 32 bytes of data:
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119
Reply from 142.250.207.68: bytes=32 time=3ms TTL=119

Ping statistics for 142.250.207.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms
```

2. Tracert.

2.tracert

```
C:\Windows\system32>tracert google.in

Tracing route to google.in [142.250.207.68]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.16.8.1
  2     1 ms     1 ms     2 ms  137.97.163.129
  3     *        *        *     Request timed out.
  4     4 ms     4 ms     5 ms  172.16.12.64
  5     3 ms     3 ms     3 ms  172.16.12.64
  6     5 ms     4 ms     4 ms  72.14.217.252
  7     5 ms     3 ms     4 ms  142.251.227.213
  8     3 ms     4 ms     3 ms  142.251.55.205
  9     3 ms     3 ms     3 ms  hkg12s32-in-f4.1e100.net [142.250.207.68]

Trace complete.
```

3. ipconfig

3.ipconfig

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::bc98:2f44:bfbf:5a6e%3
   IPv4 Address. . . . . . . . . . . : 172.16.9.19
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

4. netstat

4.netstat

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.9.19:58167      lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.9.19:58403      lcmaaa-ax-in-f5:https  ESTABLISHED
  TCP    172.16.9.19:58444      sd-in-f100:https       TIME_WAIT
  TCP    172.16.9.19:58446      4.213.25.240:https     ESTABLISHED
  TCP    172.16.9.19:58451      a104-86-188-168:https  CLOSE_WAIT
  TCP    172.16.9.19:58453      pnmaaa-ay-in-f10:https  TIME_WAIT
  TCP    172.16.9.19:58454      4.213.25.241:https     ESTABLISHED
  TCP    172.16.9.19:58455      pnmaaa-am-in-f10:https  TIME_WAIT
  TCP    172.16.9.19:58456      pnmaaa-ay-in-f10:https  ESTABLISHED
  TCP    172.16.9.19:58457      pnmaaa-bb-in-f14:https  ESTABLISHED
```

5. arp -a

5.arp -a

```
C:\Windows\system32>arp -a

Interface: 172.16.9.19 --- 0x3
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.11.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

6. Host name.

6.. Hostname

```
C:\Windows\system32>hostname
DESKTOP-0M11URO
```

7. Path ping

7 Path ping

```
C:\Windows\system32>pathping google.in

Tracing route to google.in [142.250.207.68]
over a maximum of 30 hops:
  0  DESKTOP-0M11URO [172.16.9.19]
  1  172.16.8.1
  2  137.97.163.129
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-0M11URO [172.16.9.19]
                                    0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  172.16.8.1
                                    0/ 100 =  0%   |
  2    1ms     0/ 100 =  0%     0/ 100 =  0%  137.97.163.129

Trace complete.
```

8. Route print.

```
C:\Windows\system32>route print
===========================================================================
Interface List
  3...00 27 0e 13 ee 92 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.8.1    172.16.9.19     291
        127.0.0.0        255.0.0.0         On-link     127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link     127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link     127.0.0.1     331
       172.16.8.0    255.255.252.0         On-link    172.16.9.19     291
      172.16.9.19  255.255.255.255         On-link    172.16.9.19     291
     172.16.11.255  255.255.255.255         On-link    172.16.9.19     291
        224.0.0.0        240.0.0.0         On-link     127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link    172.16.9.19     291
  255.255.255.255  255.255.255.255         On-link     127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link    172.16.9.19     291
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.8.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  3    291 fe80::/64                On-link
  3    291 fe80::bc96:2f44:bfbf:5a00/128
                                    On-link
  1    331 ff00::/8                 On-link
  3    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

Result:

Thus, the basic networking commands ping and tracert in windows were studied and executed successfully.

27/7/26 Experiment - 2

Aim:

To configure an ethernet connection by using nm CLI.

Procedure:

1). List the network manager connection by using nm CLI.

2) Create a profile:-

⇒ nm CLI connection add con-name <name> if name <device name> type ethernet.

3. Display the current setting of the connection nm CLI connection show.

4. Configure the IPV4 setting to use DHCP server.

5. configure the IPV4 settings nm CLI connection modify <connection name> IPV4 method auto.

6. Activate the profile.

nm CLI connection up network - LAN

Verification.

1. Display IP Setting of the NIC IP address show enpulse.

2. Display the IPv4 default Gateway: # ip route show default

3. Display the IPv4 default Gatew.. # ip route show default.

4. Display DNS Settings: # cat /etc/resolve.conf

5. use ping to check if this host can send packets to other.

Result:

Hence an ethernet connection was configuration using nmcli.

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14/14

3/8/26

Experiment - 3

Aim:

To study the different types of network panels.

a) Understand different types of network Cabelt.

1. Unshield twisted pair (UTP) Cable.
2. Shielded twisted pair (STP) cable
3. Coaxial Cable.
4. Fibre optic cable

Cable type	Category	Maximum Data Transmission	Advantages/ disadvantages	Application
UTP	Cat - 3 Cat - 5 Cate - 5e	10 bps upto 100Mbps 1Gbps	Advantages: *Cheaper in cost *Easy to install Disadvantages: *more prone to Electrons	10 Base -T Element fast Ethernet
STP	Cat 6,6a Cat 7	10GBPS 10GBPS	Advantages * Shielded Disadvantages. *Expensive	Gigabit Ethernet 10G Ethernet (100 m)

co-axial Cable	FG-6 RG-59 RG-11	10 - 100 mbps	Advantages: * High bandwidth * Immune to Pnface Disdvarages: * Low loss * Limited distrare	Speed of Signal is 500 mm Television
Fibre optic Cable	Signal in Signal multi people	100 Gbps	Advantages * High speed * High bandwidth Disdvarages. * Expensive * Requires skilled instally	maximum distrare of fibre optics Cables is around 100 metres.

Result:

Hence the different types of network cables are studied.

4/3/26 Experiment - 4

Aim:

To design, setup and configure an local area network (LAN).

To implementation of subnetting in CISO packet Tracker Simulator.

Creating a network topology:

The first step in implementation classes, IP subnetting is to create a network topology in packet tracer to create, select the new button in the top left corner.

Adding the devices:

Once we have created our network topology, we can add devices to it Hence, we will be adding routine, Clothes, and pc, to adding routine, Clothes and drag it onto the network topology.

Subnetting:

To subnet the network address of 192.168.10.24 to provide enough space for at least 5 address for end devices, the switch and the router, we can use a 127 subnet mask.

configuring the devices:

Now that we have added our devices and connected them, we can start configuring.

#enable

#configure terminal

#interface fast ethernet 0/0

#ip address {ip address}

#no shutdown

#exit.

Result:

The subnetting in cisco packet tracer is implemented successfully.

16/6/26 Experiment - 5

Aim:

To design a single topology and configure with routers, two signals switches and PCs using Cisco packet tracer.

Build the topology:

* Add End devices navigate to end devices in the bottom - left palette. Drag from PCs into the logical workspace place PC0 and PC1 on the left.

* Add switches: Go to network devices. Switches choose the CISCO 2960 switch Drag two switches onto the the workspace.

* Cable the device: Select the connection icon and choose the copper straight - through cable.

* Connect PC1 (Fast Ethernet 0) to switch 0 (fast Ethernet 0/2)

* Select static and fill in the following details:

* IPv4 address: 192.168.10.10

* Subnet mask : 255.255.255.0

* Default gateway : 192.168.0.1

PC1 configuration:

* IP Address: 192.168.10.11

* Subnet mask: 255.255.255.0

* Default Gateway: 192.168.10.1

Verify connectivity:

* Click on PC0 go to the and open the CP.

* Run a command - the ping to yours local gateway to configure the local

* ping 192.168.10.1

* Test inter - subnet routing cursors the network.

Result.

The network topology is operational ping out - due to address resolution protocol

13/8/26 Experiment - 6.

Aim:

To customize a modular switch in cisco packet tracer by installing fast Ethernet and gigabit filter network module, connect it to PCs and a router, and verify network connectivity

Procedure:

* open Cisco packet tracer
* from network Devices - Switches, Selected PT - EMPTY and place at the workspace.
* Add two PCs and one CISCO 2911 router to the workspace
* open the switch and go the Physical tab.
* Turn off switch before installing any modules.
* From the modules itself.
* Insert PT - switch NM - IFE

into slots.

* Insert the HWIC-1GE SFP module into an empty slot

* Check the iface using:

Router # Show ip interface brief

Verify the fiber interface is up/up

* Test connectivity using router ping 192.168.10.2.

Result:

The 2960-24PT-EMPTY was successfully customised with fast Ethernet modular 2911 router was also filled with a fibre compatible module.

11/3/24 Experiment - 7

Aim:

Write a program to implement flow control at data link layer using sliding window the flow of frames from one node to another.

Procedure:

* Input window size from the user.
* Create a frame with the following fields.
* Send the frames and wait for the acknowledgement from the receiver.
* Reader a file called Receiver Buffer.
* Check ACK field for the Acknowledgement number.
* If the Acknowledgement from the receiver number is expected, send new set of frames accordingly, Is received, resend the frames accordingly.

Create a receiver file with the following feature.

1. Reader a file called Sender_Buffer
2. Check the frame number.
3. If the frame numbers are expected Write appropriate ACK no in the Receiver_Buffer.

Result

This program implements a sliding window error-control protocol using file buffers to exchange frames with ACK/NACK sequence verification.

Exno: 8

17/8/26 NMAP to discover live Hosts using Nmap Scans on tryHackMe Platform.

Aim:
To discover live Hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on tryHackMe platform.

NMAP:
Nmap stands for network Mapper. It is open-source network Subnetting Scanning tool used to identify live hosts & network services.

Sends a packet with the following

From
Computer 1

To
Computer 1

packet type
arp-request

Data
Computer 6.

From computer 1

To computer 1

Packet type ARP Request.

2. Did computer receive the ~~ARP~~ ARP request ?

Send a packet with the following

From

computer 4

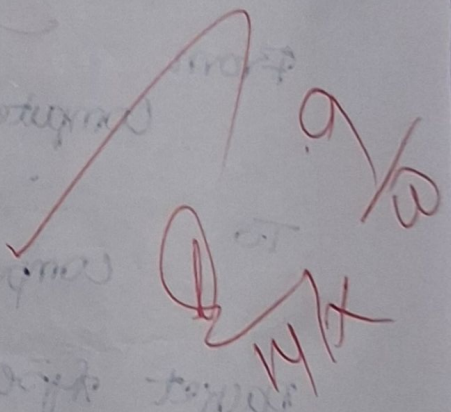
To Computer 4

Packet type

arp - request

Data

computer 6.



Result.

Thus to discover live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on TryHackMe platform is executed successfully.

Ex:no:9
24/3/26

Implementation of Subnetting in Cisco packet Tracer Simulator

Aim:

To implement Subnetting in Cisco Packet tracer Simulator.

Creating a network topology:

The first step in implementing classless IP Subnetting is to create a network topology in packet tracer.

Subnetting:

To subnet the network address of 192.168.1.0 to provide enough space for at least 5 address for end devices, the switch and the router, we can use a 127 subnet mask. This will give us 8 subnets with 30 host address each.

Result:

Thus an experiment to implement subnetting in Cisco Packet tracer is executed successfully.

Exno:11 Building Anonymous FTP Scanner.
31/3/25 using ftplib module

Aim:

To build anonymous FTP Scanner using python's ftplib module to detect open FTP servers that allow anonymous login.

Procedure:

1. Import the ftplib module in python.
2. Define a function to connect to the target FTP server on port 21.
3. Attempt to log in using the anonymous username.
4. Check whether the server accepts anynmous username.
5. Display the welcome message if login is successfully.
6. Handle login, connection, and timeout errors using exception handling

8. Run the function with the target FTP Server / IP and observe the result.

Code:

```
Import ftplib
def check_anonymous_login(hostname):
    try:
        ftp = ftplib.FTP(hostname, timeout=5)
        welcome_msg = ftp.login('anonymous',
        'anonymous@gmail.com').
        print(f"[+] {hostname} allows
        anonymous login").
        print(f" Welcome message: {
            welcome_msg}") ftp.quit()
        return True
    except ftplib.error_perm as e:
        print(f"[-] {host_name}:
            anonymous login
            denied ({e})")
        return False;
```

Result:

The anonymous FTP Scanner was Implemented Successfully

Ex.no:12 End - End Communication at transport layer.

15/9/24

Aim:

To implement an echo client - server program using TCP sockets in Python.

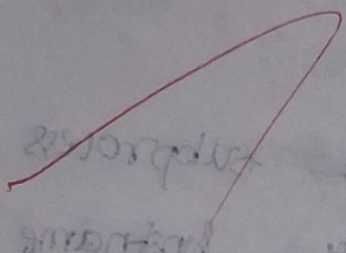
Procedure:

* Create a server socket and bind it to an IP address and port number.
* Listen for client connection and accepts a connection.
* Receive a message from the client.
* Send the same message back to the client.
* Display the echoed message the terminate.

Code. server.py:

```
import socket
s = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
s.connect(("127.0.0.1", 5000))
```

```
msg = input ("Enter message:")
s.sendall (msg.encode())
data = s.recv (1024)
print ("Echo from Server:", data.decode
                                    ())
s.close().
```



Result:

Thus, the echo client - server program was implemented using TCP sockets.

Ex:no:13 Implement your own ping program.
15/9/26

Aim:

To implement a simple ping program to check the connectivity of romote host.

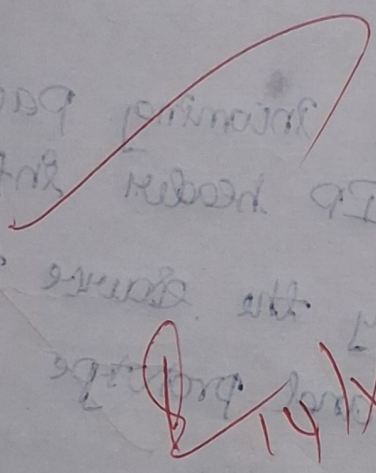
Proledure:

* Import the subprocess module.
* Obtain the hostname or IP address from the user.
* Execute the system ping command using python.
* Send ICMP echo requests to the specified host.
* Display the connectivity status and response time information.

Code:

```
import subprocess
import platform
host = input("Enter hostname or IP address:")
```

```
param = "-n" if
    platform.system().lower() == "windo
else
    "-c"
try:
    result = subprocess.run(
        ["ping", param, "4", host],
        text=True,
        timeout=15
    )
print(result.stdout)
```



Result:

Thus, a simple ping program was implemented successfully to check network connectivity.

Ex: no 4. Implement packet Sniffing.
22/9/26 using RAW Sockets.

Aim:
To capture and and analyse netw packets using RAW sockets in Python.

Procedure:
* Import the Socket module.
* Create a RAW Souket using th appropriate protocol
* Bind the Souket to the netw interface if required by the operating System.
* capture incoming packets and extract the IP header information.
* Display the source IP, destination IP and protocol type.

python:
import Sokket
Import Socket.
try:
s=socket.Saket(

```
    Socket.AF_INET,
    Socket.SOCK_RAW,
    Socket.IPPROTO_IP
)
s.bind(("0.0.0.0", 0))
while True:
    print("Socket error:", e)
```

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Result:

Thus network packet were captured and their basic IP headers information was displayed using RAW Sockets.

EXNO:15 Analyse various types of servers
26/9/26. using the Webalizer Tool.

Aim:

To analyse web server log files and generate statistical report using the Website tool.

Procedure:

* install webalizer on a compatible Linux System.

* Locate the Web server access log file.

* configure webalizer with the log file path and output directory

* Execute webalizer to generate HTML reports.

* Open the generated report in browse and analysis visits, requests, bandwidth, and referring sites.

Bash:

```
Sudo apt update
Sudo apt install webalizer
Sudo mkdir -p /var/www/html
Sudo webalizer
    -o/var/www/html/webalizer.
```

8/14/x 10/10

Result:

Thus, web server access logs were analyzed using webalizer and statistical reports were generated to study server traffic and usage.